



BASES DE DATOS AVANZADAS

Classroom: Bases de Datos Avanzadas



12 DE MAYO DE 2026

carlos.herrera@upa.edu.mx

<https://github.com/ISC-UPA/2026-2-TIID5-DB2>

Contenido

- 1. S01..... 2
 - 1.1. Referencias 2
 - 1.2. Technological Requirements 2
 - 1.3. Relaciones..... 2
- 2. S02..... 3

1. S01

1.1. Referencias

<https://www.w3schools.com/mysql>

<https://conclase.net/mysql>

1.2. Technological Requirements

VS Code <https://code.visualstudio.com/Download>

Extensions: **Database Client JDBC**

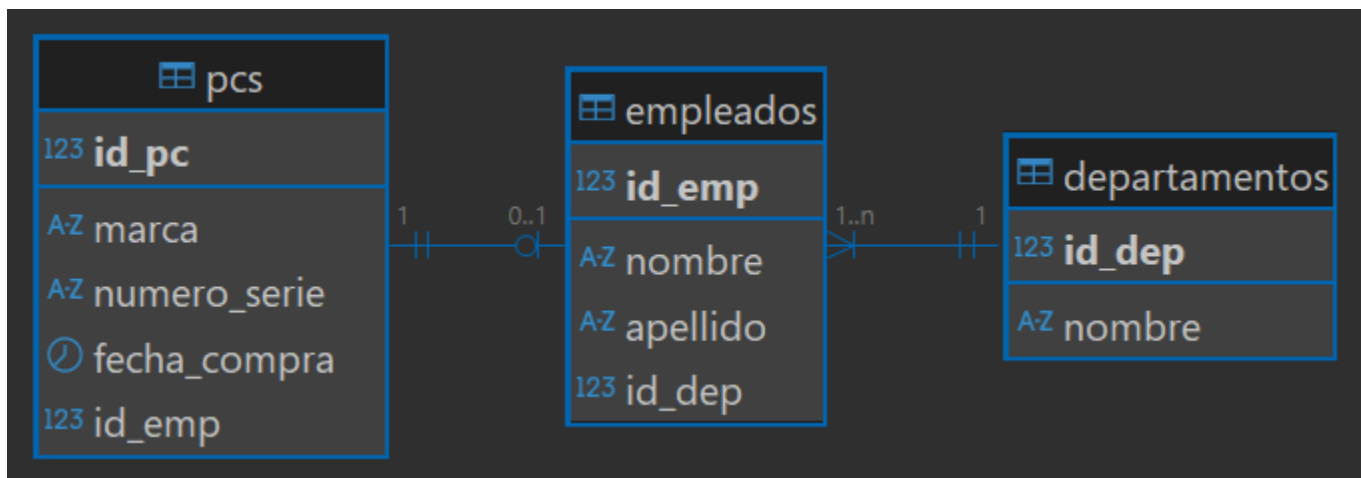
Extensions: **MongoDB for VS Code**

XAMPP

MongoDB

PowerBI

1.3. Relaciones



```
-- Crear tabla de departamentos
CREATE TABLE departamentos (
    id_dep INT PRIMARY KEY AUTO_INCREMENT,
    nombre VARCHAR(50) UNIQUE NOT NULL
);

-- Crear tabla de empleados, debe pertenecer a un departamento
CREATE TABLE empleados (
    id_emp INT AUTO_INCREMENT,
    nombre VARCHAR(50),
    apellido VARCHAR(50) NOT NULL,
    id_dep INT not null,
    CONSTRAINT pk_empleados PRIMARY KEY (id_emp),
    FOREIGN KEY (id_dep) REFERENCES departamentos(id_dep)
);

-- Crear una tabla de pcs,
-- cada pc puede estar asignada a un empleado
-- cada empleado debe tener una pc asignada.
```

```

CREATE TABLE pcs (
    id_pc INT PRIMARY KEY AUTO_INCREMENT,
    marca VARCHAR(50),
    numero_serie VARCHAR(20) NOT NULL,
    fecha_compra DATE,
    id_emp INT,
    UNIQUE INDEX unique_serie (numero_serie ASC) VISIBLE
);

create unique index unique_emp on pcs(id_emp);

alter table pcs
add constraint fk_pcs_emp foreign key (id_emp) references empleados(id_emp);

-- Insertar datos en la tabla de departamentos
INSERT INTO departamentos (nombre) VALUES ('Ventas');
INSERT INTO departamentos VALUES (null, 'Informatica');
INSERT INTO departamentos VALUES (null, 'Recursos Humanos');

-- Insertar datos en la tabla de empleados
INSERT INTO empleados (nombre, apellido, id_dep) VALUES ('Jesus', 'Pérez', 1);
INSERT INTO empleados VALUES (null, 'María', 'Gómez', 2);
INSERT INTO empleados VALUES (null, 'Jose', 'Franco', 2);

-- Insertar datos en la tabla de pcs
INSERT INTO pcs (marca, numero_serie, fecha_compra, id_emp) VALUES ('Dell', 'SN12345', '2022-01-15', 1);
INSERT INTO pcs VALUES (null, 'HP', 'SN54321', '2023-03-10', 2);
INSERT INTO pcs VALUES (null, 'ASUS', 'SN67890', '2024-10-15', null); -- PC sin asignar

UPDATE pcs SET id_emp = 2 WHERE id_pc = 1;
-- Un Empleado una PC

```

-- DDL Language Definition Data = Crear Tabla, Vista, Procedimiento Almacenado, Función, Índice, Esquema, Base de Datos etc.

-- DML Language Data Manipulation Data = Insertar, Actualizar, Eliminar, Seleccionar etc.

-- DCL Language Definition Control Data = Conceder, Revocar etc.

-- TCL Language Definition Transaction Control Data = Commit, Rollback, Savepoint etc.

Funciones: select salary, codigo(salary) from employees;

Procedimientos: exec sp_calcularImpuesto @deptoid = 101;

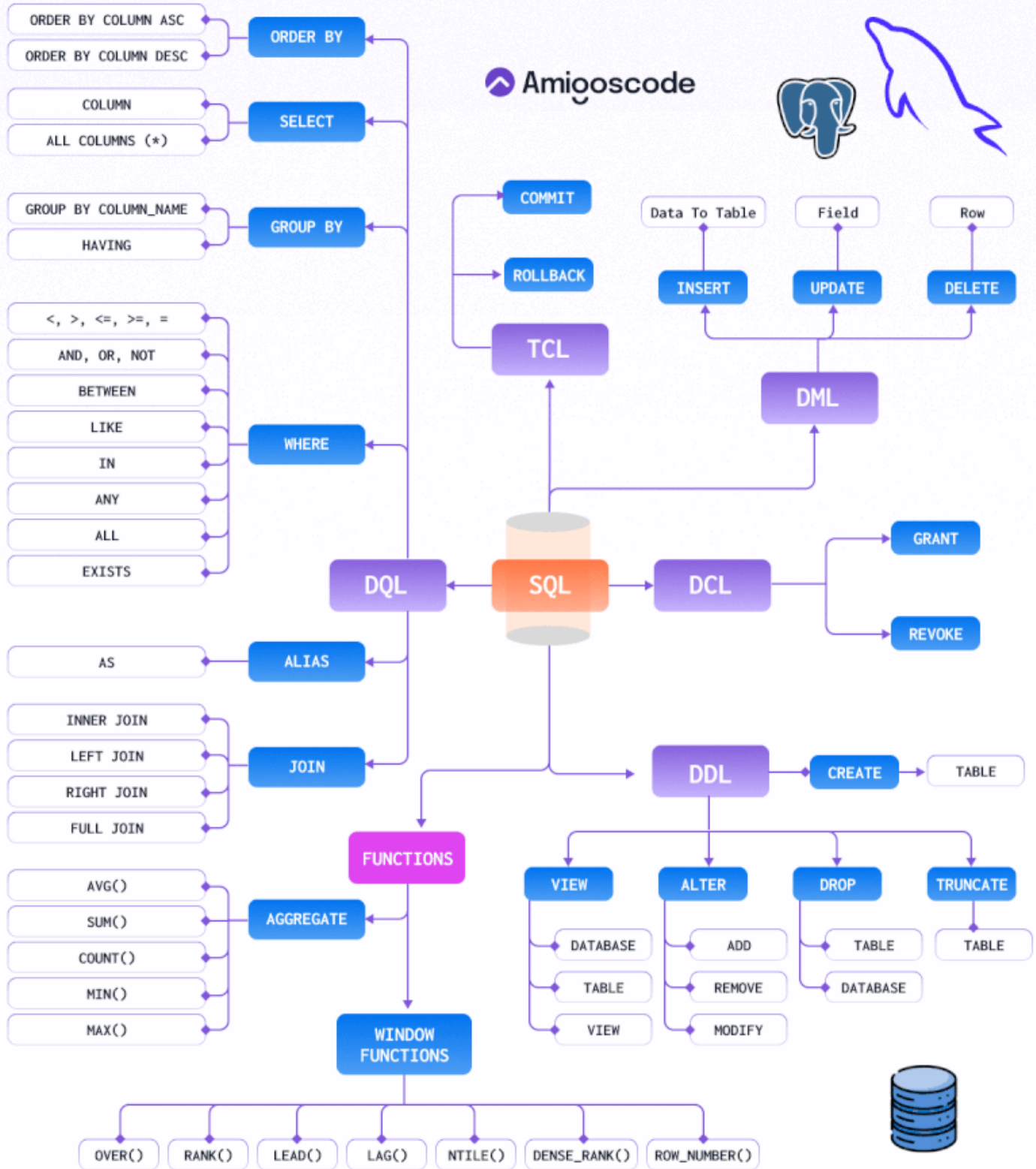
2. S02

Select	fields
from	tables
where	filters
	logical operators: not and or
	relational operators: >, <, >=, =, <>, !=
	some words: like, between, in, is null, is not null
group by	functions: min, max, count, sum, avg, std
having	filters with functions
order by	fields [desc]

```
-- JOIN
SELECT table1.column, table2.column
FROM table1
    [NATURAL JOIN table2] |
    [JOIN table2 USING (column_name)] |
    [inner JOIN table2
        ON (table1.column_name = table2.column_name)] |
    [LEFT|RIGHT|FULL OUTER JOIN table2
        ON (table1.column_name = table2.column_name)] |
    [CROSS JOIN table2];
```

SQL COMMANDS

Amigoscode



Amigoscode